Huffman Coding

"DATA ANALYTICS AND INTELLIGENCE LABORATORY"

Apply Huffman Coding algorithm.

S-1:-

Letter Count

A[7], L[4] N[4] T[4] E[3] I[3] C[2] D[2] O[2]

R[2] Y[2] B[1] G[1] S[1]

S-2:- Ascending order

B[1] - G[1], S[1], C[2], D[2], O[2], R[2], Y[2], E[3], I[3],

L[4], N[4], T[4], A[7]

S-3:-

Apply Huffman algorithm

B[1] - G[1] - S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - E[3] - I[3]

L[4] - H[4] - T[4] - A[7]

-1:-

S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - DG[2] - E[3] - I[4] -

L[4] - N[4] - T[4] - A[7]

-2:-

Combine S[1] & C[2] to SC[3]

D[2] - O[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4]

T[4] - A[7]

3:-

R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4]

DO[4] - A[7]

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I-4:-

BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - WO[4] - RY[4] - A[7]

I-5: Combine BG[2] & E[3]

I [3] - SC[3] - L[4] - N[4] - T[4] - WO[4] - RY[4] - BGE[5] - A[7]

I-6:-

L[4] - N[4] - T[4] - WO[4] - RY[4] - BGE[5] - ISC[6] - A[7]

I-7:-

T[4] - WO[4] - RY[4] - BGE[5] - ISC[6] - A[7] - LN[8]

I-8:- Combine T[4] & WO[4]

RY[4] - BGE[5] - ISC[6] - A[7] - LN[8] - TWO[8]

I-9:-

ISC[6] - A[7] - LN[8] - TWO[8] - RYBGE[9]

I-10

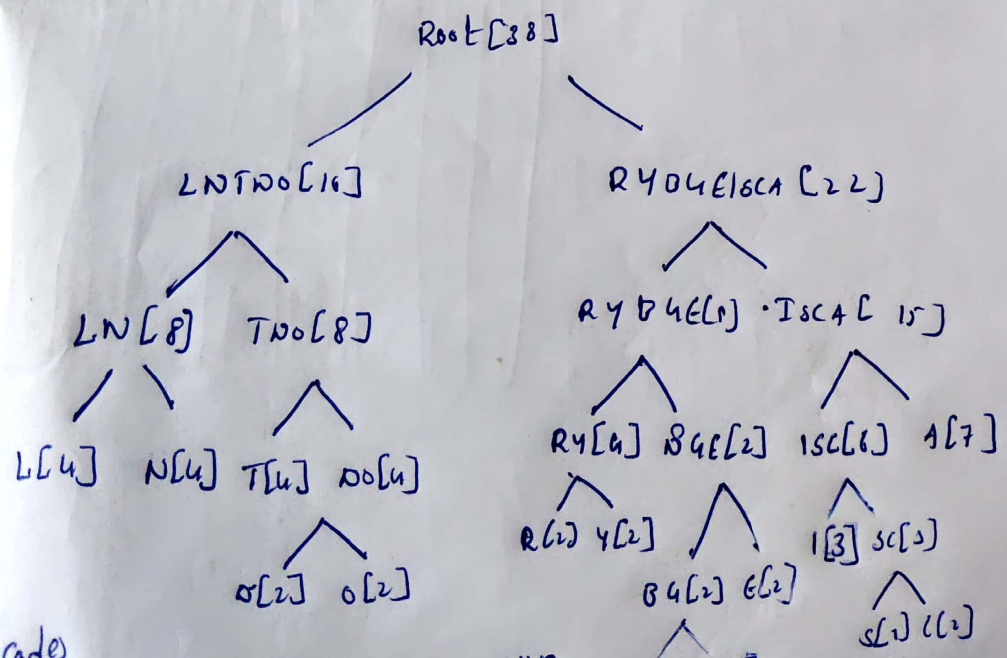
LN[8] - TWO[8] - RYWGE[9] - ISEA[13]

I-11

RYBGE~~ISCA~~[9] - ISCA[13] - LNTWO[16]

I-12

LN·TWO[16] - RYBGEISCA[22]



Codes

A = 111
L = 000
N = 001

T = 010
E = 1011
I = 1100

W = 0110
O = 0111
R = 1000
Y = 1001
B = 10100

G = 10101
S = 11010

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